

Supplementary File 1: Search Criteria for Evaluated Libraries

MechaScreener: Large Language Model-Based Automated Screening for Systematic Reviews and Research

Introduction

This supplementary file details the exact screening criteria provided to the MechaScreener Large Language Model (LLM) for each of the 15 reference libraries analysed in the main text. The libraries are divided into 5 development libraries and 10 evaluation libraries (5 focusing on Randomized Controlled Trials [RCTs] and 5 focusing on non-RCTs). Criteria were structured either using the Population, Intervention, Comparator, Outcome, and Type of study (PICOT) format, or provided as general free-text inclusion/exclusion rules. Only the “include” criteria are listed below; fields left blank in the system are omitted.

Contents

1	Development Libraries	2
1.1	Antibiotic Prescribing and Telehealth	2
1.2	Long COVID	2
1.3	Natural History of Primary Care	2
1.4	Non-Drug Interventions	3
1.5	Salt Substitution	3
2	Evaluation Libraries (RCTs)	3
2.1	Balneotherapy for Chronic Venous Insufficiency	3
2.2	Calcium and Vitamin D for Increasing Bone Mineral Density in Premenopausal Women	4
2.3	Methylxanthine for the Prevention and Treatment of Apnea in Preterm Infants	4
2.4	Phosphodiesterase 5 Inhibitors (PDE5i) for the Treatment of Raynaud’s Phenomenon	4
2.5	Topical and Oral Steroids for Otitis Media with Effusion (OME) in Children	5
3	Evaluation Libraries (Non-RCTs)	5
3.1	Covid-19 in Schools	5
3.2	Diabetes TB	5
3.3	Falls Prevention	5
3.4	Fluoride and Fluorosis	6
3.5	Sanitation for Preventing Diarrhea	6

1 Development Libraries

1.1 Antibiotic Prescribing and Telehealth

Format: PICOT

Population:

adult or paediatric patients with a history of a community acquired acute infection (respiratory, urinary, or skin and soft tissue)

Intervention:

any type of telehealth consultations (phone or video based)

Comparator:

face to face consultations

Outcome:

the number of antibiotic prescriptions in each type of consultation, follow-up visit rates, testing rates or number of samples sent to the laboratory, any reported adverse events

Type of study:

randomized trials or observational studies

1.2 Long COVID

Format: PICOT

Population:

people of all ages who were eligible to receive a covid-19 vaccine

Intervention:

any dose of a covid-19 vaccine

Comparator:

no vaccination, an active non-covid-19 vaccine control (eg, influenza vaccine), or placebo

Outcome:

patients with a diagnosis of long covid, prevalence of individual symptoms of long covid, such as prolonged fatigue, shortness of breath, cognitive difficulties, and loss of sense of smell

1.3 Natural History of Primary Care

Format: General/Other

Eligibility criteria were: clinical practice guidelines produced by an international or national organisation involved in the publication of guidelines, and contained clinical recommendations on the management of acute infections that are commonly seen in primary care and may be managed with antibiotics (including acute respiratory infections (ARIs), lower urinary tract infections (UTIs), skin and soft tissue infections (SSTIs), conjunctivitis).

Inclusion criteria:

1. Acute respiratory infections including acute otitis media, sore throat, pharyngitis, tonsillitis, common cold, sinusitis, laryngitis, acute cough, acute bronchitis, bronchiolitis middle ear inflammatory conditions
2. Rhinosinusitis, rhinitis, group A streptococcal infections, streptococcal pharyngitis
3. Influenza-like illnesses including seasonal influenza

4. Lower urinary tract infections, cystitis, acute bladder infection, recurrent UTI
5. Skin and soft tissue infections, cellulitis, acne, impetigo, boils, carbuncles, ecthyma
6. All forms of conjunctival conditions
7. Management of croup
8. Urinary tract infections in non-pregnant women
9. Acute exacerbation of COPD

1.4 Non-Drug Interventions

Format: General/Other

Include a study if it meets ALL of the following criteria:

1. **Study Type:** The study must be a systematic review or a scoping review.
2. **Topic:** It must identify the barriers or enablers of using non-drug interventions (e.g., diet, exercise, psychological support, self-management).
3. **Condition:** The intervention must be for the prevention or management of a chronic disease.
4. **Setting:** The setting must be primary care, where a clinician referred the patient to the intervention.
5. **Data Source:** The review must be based on primary studies that are mostly qualitative or mixed-methods.

1.5 Salt Substitution

Format: PICOT

Population:

adults (aged ≥ 18 years)

Intervention:

studies of salt substitution interventions (that is, salt substitute intake compared with regular salt intake) delivered through advice and recommendation from a clinician or researcher, or the provision of salt substitute product

Comparator:

studies that used no intervention or regular table salt as a comparator

Outcome:

mortality, MACE (for example, any composite of myocardial infarction, revascularization, stroke, and heart failure), serious adverse events (for example, renal failure or severe hyperkalemia), SBP and DBP

Type of study:

randomized trials

2 Evaluation Libraries (RCTs)

2.1 Balneotherapy for Chronic Venous Insufficiency

Format: PICOT

Population:

people with venous insufficiency

Intervention:

patients treated with balneotherapy

Outcome:

Edema OR Leg Ulcer OR Pain OR Skin Pigmentation OR Adverse Events OR Quality of Life

Type of study:

randomized trials

2.2 Calcium and Vitamin D for Increasing Bone Mineral Density in Pre-menopausal Women

Format: PICOT

Population:

Adult women

Intervention:

People given Calcium or Vitamin D

Outcome:

Increased bone mineral density or reduction in fractures or quality of life improvements or adverse events

Type of study:

randomized trials

2.3 Methylxanthine for the Prevention and Treatment of Apnea in Preterm Infants

Format: PICOT

Population:

Preterm infants

Intervention:

People given Methylxanthine or Aminophylline or Theophylline or Caffeine

Outcome:

Positive Pressure Therapy or Chronic Lung Disease or Apnea or Death or Neurodevelopmental Disorder

Type of study:

randomized trials

2.4 Phosphodiesterase 5 Inhibitors (PDE5i) for the Treatment of Raynaud's Phenomenon

Format: PICOT

Population:

Adults with Raynaud's Phenomenon

Intervention:

People given Phosphodiesterase 5 Inhibitors

Outcome:

Frequency of attacks or Duration of attacks or Severity of attacks or Raynaud's condition score or Pain or Adverse Events OR Patient Improvement

Type of study:

randomized trials

2.5 Topical and Oral Steroids for Otitis Media with Effusion (OME) in Children

Format: PICOT

Population:

Children with Otitis Media with effusion

Intervention:

People given steroid therapy

Outcome:

Hearing or Quality of Life or Adverse Events or persistence of otitis media

Type of study:

randomized trials

3 Evaluation Libraries (Non-RCTs)

3.1 Covid-19 in Schools

Format: PICOT

Population:

people involved in school settings

Intervention:

measures to reduce transmission of COVID-19

Outcome:

“transmission-related outcomes” OR “healthcare utilisation” OR “economic outcomes” OR “social outcomes” OR “ecological outcomes”

3.2 Diabetes TB

Format: PICOT

Population:

patients with diabetes

Outcome:

incidence of tuberculosis

Type of study:

cohort studies

3.3 Falls Prevention

Format: PICOT

Population:

patients over 45 years old

Intervention:

“fall prevention” OR “exercise” OR “safety education” OR “nutrition therapy”

Comparator:

usual care

Outcome:

“incidence of falls” OR “hospitalisations for falls” OR “fall-related injuries” OR “cost of healthcare associated with falls”

Type of study:

“randomised controlled trials” OR “step-wedge designs” OR “non-RCTs”

3.4 Fluoride and Fluorosis

Format: PICOT**Population:**

children exposed to topical fluoride

Intervention:

topical fluoride

Comparator:

“non-topical fluoride treatments” OR “placebo” OR “no intervention”

Outcome:

prevalence of fluorosis

Type of study:

“randomised controlled trials” OR “observational studies” OR “cohort studies” OR “case control studies” OR “cross-sectional surveys”

3.5 Sanitation for Preventing Diarrhea

Format: PICOT**Population:**

“children” OR “adults”

Intervention:

sanitation measures

Comparator:

“open defecation” OR “current practice”

Outcome:

prevalence of diarrhoea

Type of study:

“randomised controlled trials” OR “quasi-RCTs” OR “non-RCTs” OR “controlled before and after studies” OR “matched cohort studies”